

TABLE OF CONTENTS

TABLE OF CONTENTS	1
1- SERVICE A CIRCUIT BOARD	2
2- REPLACE AN SCR	3
3- REPLACE MAIN CIRCUIT BREAKER (A8A1)	5
4- BYPASS SWITCH	8
5- A TERMINAL BLOCK (MODELS 46-317099P1/P2)	10
6- SURGE SUPPRESSOR (MODELS 46-317099P1/P2)	14
7- SURGE SUPPRESSOR (MODELS 46-317099P8 THRU P12)	15
8- SHUNT MODULE (MODELS 46-317099P1/P2)	17
REVISION HISTORY	19

Description - The Compact PDU is a single-bay cabinet and the Standard PDU is a two-bay cabinet. Procedures for the Compact PDU do not apply to the Standard PDU.

1- SERVICE A CIRCUIT BOARD



LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.
3. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
4. Turn main circuit breaker OFF.
5. Turn load panel circuit breakers OFF.
6. Measure input test points (A8) for 0 VAC before proceeding.
7. Disconnect connectors on circuit board. Remove board from standoffs.
8. Install replacement board on standoffs. Reconnect connectors.
9. If a logic board is replaced, perform prestartup and basic check on logic board. Refer to procedure for Compact PDU, Section 5 - Startup Checks and Section 7 - General Checks.
10. Perform functional check. Refer to procedure for Compact PDU, Section 8 - Regulator Checks.
11. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
12. Unlock facility circuit breaker and turn on.

13. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

14. Press Emer. Stop Reset switch, and reset following circuit breakers:

a. SIGNA HORIZON:

Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet), GRAM Cabinet (if cabinet installed), and RF Amplifier Cabinet

SIGNA ADVANTAGE:

a. Gradient Amplifier Cabinet, RF Amplifier Cabinet and Penetration Cabinet circuit breakers.

b. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.

15. Turn load panel circuit breakers ON.

16. Close outer door.

2- REPLACE AN SCR



LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)

2. Open outer door of Compact PDU.

3. Turn main circuit breaker OFF.

4. Measure input test points (A8) for 0 VAC before proceeding.

5. Turn all load panel circuit breakers OFF.

6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Disconnect transformer tap lead from SCR. See Illustration L2735A below.

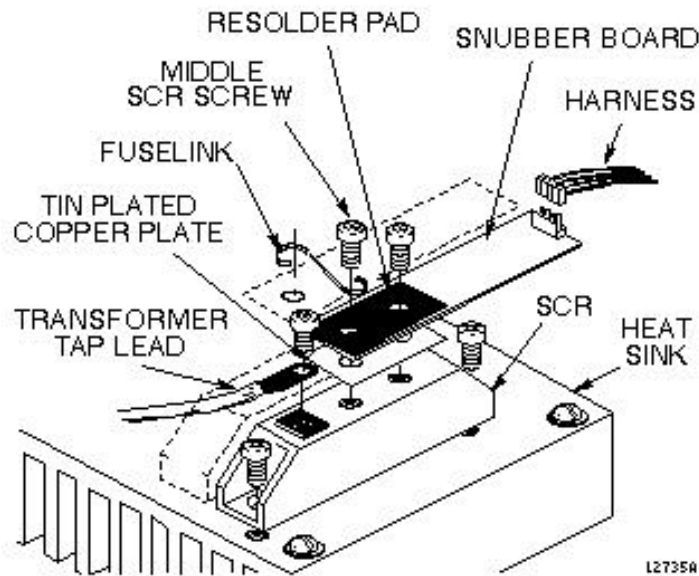


ILLUSTRATION L2735A
SCR ASSEMBLY

8. Remove fuselinks and snubber board from SCR. Remove SCR from heat sink.
9. Spread a thin film of heat-sink compound on new SCR and mount on heat sink.
10. Attach snubber board to SCR. Attach fuselinks to snubber board. Torque SCR screws to 30 lb in (34 Nm).
11. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center of right edge of door.
12. Perform basic check. Refer to procedure for Compact PDU General Checks, Section 1 - General Checks.
13. Perform functional check. Refer to procedure for Compact PDU Regulator Checks, Section 1 - Regulator Check (46-317099P1/P2 & P8 Thru P11 Models).
14. Close outer door.

3- REPLACE MAIN CIRCUIT BREAKER (A8A1)

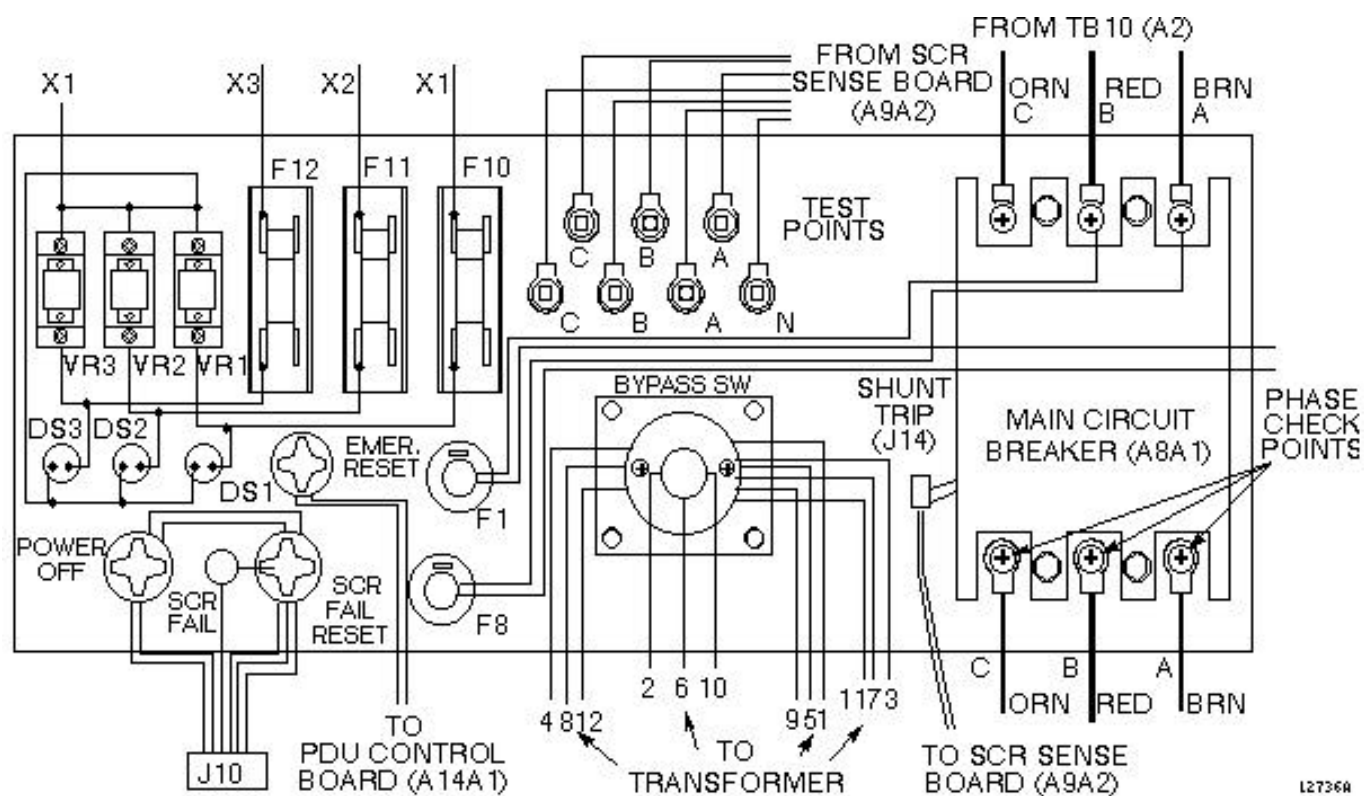


LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Turn all load panel circuit breakers OFF.
6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove main circuit breaker.
 - a. Disconnect main circuit breaker shunt trip (J14). See Illustration L2736A below or Illustration L4794A under illustration L2736A.

Note

Main circuit breaker & Bypass switch configurations are the same for P1/P2 (shown) and P3/P4.



L2736A

ILLUSTRATION L2736A
FRONT PANEL (REAR VIEW) MODEL 317099 P1/P2 SHOWN

Note

Main circuit breaker & Bypass switch configurations are the same for P1/P2 (shown) and P3/P4.

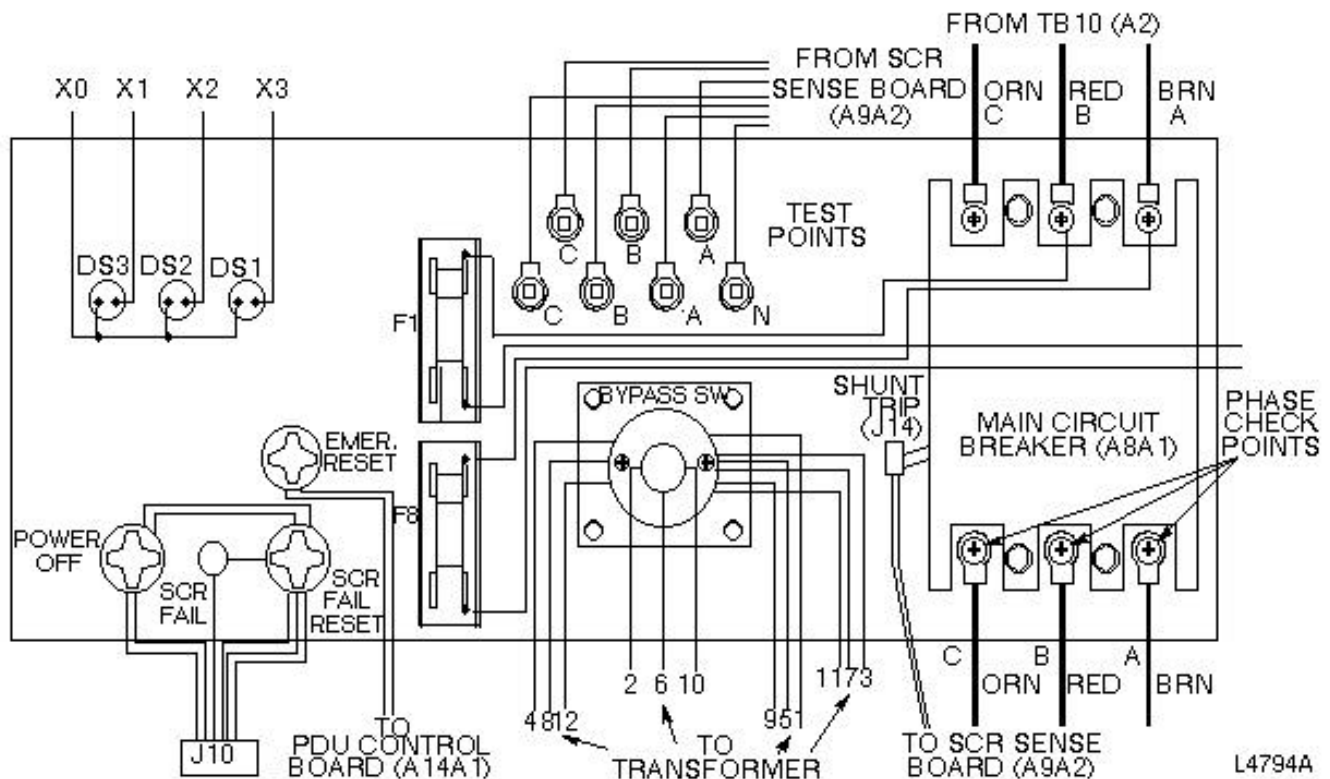


ILLUSTRATION L4794A
FRONT PANEL (REAR VIEW) MODEL 46-317099 P8 THRU P11 SHOWN

- b. Remove two small wires at top of main circuit breaker.
 - c. From front panel side, remove and keep four screws, eight washers and four nuts securing main circuit breaker to front panel and two spacer plates.
 - d. Remove six power connections to main circuit breaker. Remove main circuit breaker.
8. Replace main circuit breaker. Be sure wire color on input is the same as wire color on output!
 - a. Install six power connections to main circuit breaker. Be sure ring terminals are in place on top red and brown power connections. Torque to 25 lb in.
 - b. Holding circuit breaker in place, install four screws, eight washers and four nuts securing main circuit breaker to front panel.
 - c. Install two fuse wires at top of main circuit breaker.
 - d. Connect main circuit breaker shunt trip (J14).
9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
10. Unlock facility circuit breaker and turn on.

11. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

12. Press emer. stop reset switch and reset following circuit breakers:
 - a. SIGNA HORIZON:
Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet), GRAM Cabinet (if cabinet installed), and RF Cabinet

SIGNA ADVANTAGE:
Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet.
 - b. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.
13. Turn load panel circuit breakers ON.
14. Close outer door.

4- BYPASS SWITCH



LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.
3. Turn main circuit breaker OFF.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Turn all load panel circuit breakers OFF.

6. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
7. Remove bypass switch. See Illustration L2736A above in Section 3 or Illustration L4794A above in Section 3.
 - a. On front panel, remove and keep screw securing knob to bypass switch. Remove knob.
 - b. Remove and keep four screws securing bypass switch to front panel. Bypass switch will come free of panel.
 - c. Tag wires on bypass switch to identify where they go on the new bypass switch.
 - d. Remove wires from bypass switch.
 - e. Remove bypass switch.
8. Replace bypass switch.



Connect wires to the same location on bypass switch from which they were disconnected. Severe damage to equipment may result if wiring is incorrect.

- a. Connect wires to bypass switch. Torque nine power connections to 30 lb in (34 Nm).
 - b. Holding bypass switch in place, install four screws securing bypass switch to front panel.
 - c. Holding knob in place, install screw securing knob to bypass switch.
9. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
10. Unlock facility circuit breaker and turn on.
11. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

12. Press Emer. Stop Reset switch, and reset following circuit breakers:

a. SIGNA HORIZON:

Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet),
GRAM Cabinet (if cabinet installed), and RF Amplifier Cabinet

SIGNA ADVANTAGE:

Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet.

- b. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.

13. Turn on load panel circuit breakers.

14. Close outer door.

5- A TERMINAL BLOCK (MODELS 46-317099P1/P2)



LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.
3. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Disconnect four wires between terminal block and filter.
 - **Model 46-317099P1:** See Illustration L2737A below and refer to Table 1 under Illustration L2739A.



ILLUSTRATION L2737A

FILTER WIRING DIAGRAM (60 HZ)

TABLE 1
FILTER WIRING DIAGRAM (MODEL 46-317099P1)

WIRE LIST

WIRE	CONNECT TO
BRN-1, BRN-2	SU-1B, SU-2B
RED-1, RED-2	SU-3B, SU-4B
ORN-1, ORN-2	SU-5B, SU-6B
WHT/BLU-1	SU-1A
WHT/BLU-2	SU-4A
WHT/BLU-3	SU-6A
BRN-3	MA-A (LINE)
BRN-4	MA-A (LOAD)
RED-3	MA-B (LINE)
RED-4	MA-B (LOAD)
ORN-3	MA-C (LINE)
ORN-4	MA-C (LOAD)
WHT/BLU-4	MA-N (LINE)
WHT/BLU-5	MA-N (LOAD)

- **Model 46-317099P2:** See Illustration L2739A below and refer to Table 2 following Illustration L2739A.

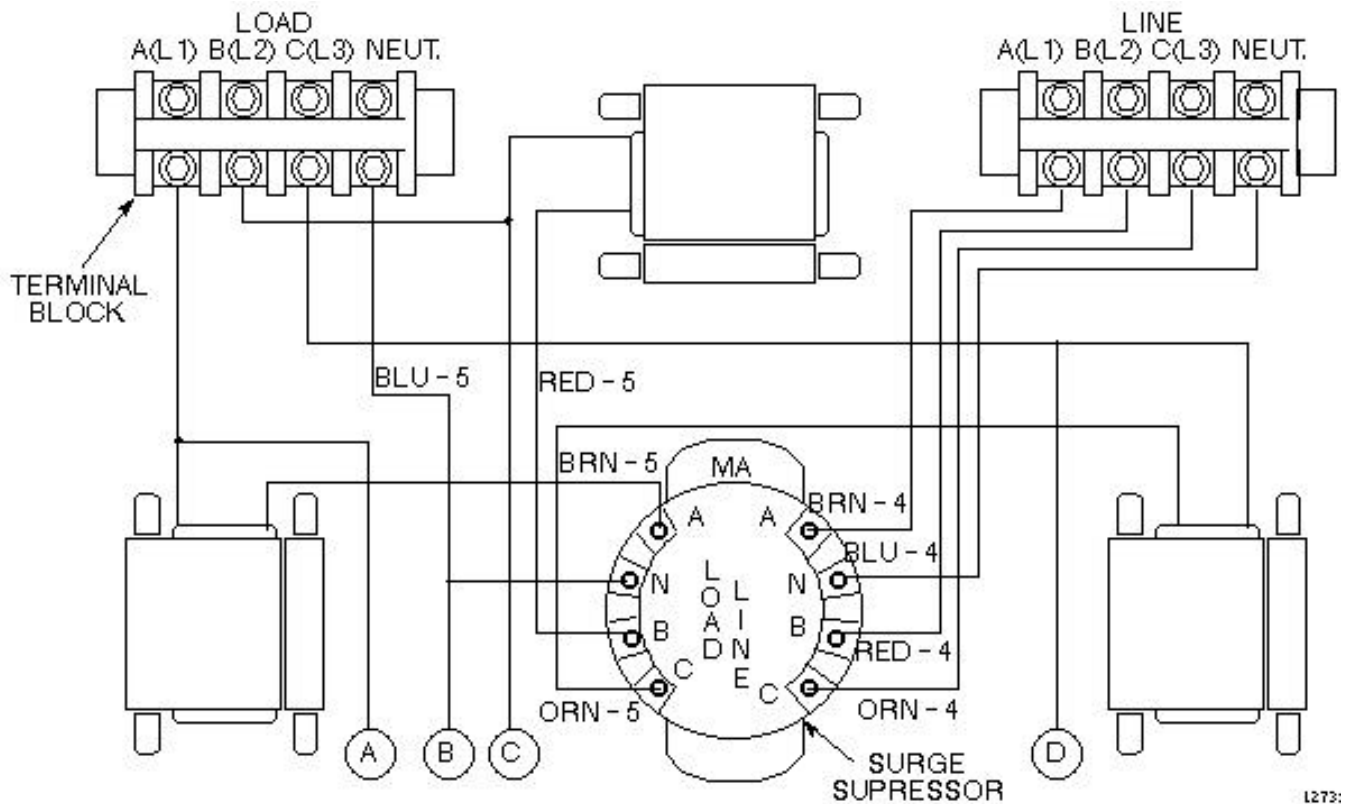


ILLUSTRATION L2739A
FILTER WIRING DIAGRAM (50 HZ)

TABLE 2
FILTER WIRING DIAGRAM (MODEL 46-317099P2)

WIRE LIST

WIRE	CONNECT TO
BRN-1, BRN-2, BRN-3	SU-1A, SU-2B, SU-3B
RED-1, RED-2, RED-3	SU-4A, SU-5B, SU-6B
ORN-1, ORN-2, ORN-3	SU-7A, SU-8B, SU-9B
BLU-1	SU-1B
BLU-2	SU-4B
BLU-3	SU-7B
BRN-4	MA-A (LINE)
RED-4	MA-B (LINE)
ORN-4	MA-C (LINE)
BLU-4	MA-N (LINE)
BRN-5	MA-A (LOAD)
RED-5	MA-B (LOAD)
ORN-5	MA-C (LOAD)
BLU-5	MA-N (LOAD)

6. Disconnect wires between terminal block and Compact PDU.
7. Remove and keep four nuts, star washers, flat washers and screws securing terminal block to filter assembly. Use a 3/8" socket for nuts. Remove terminal block.
8. Install new terminal block. Secure to filter assembly with flat washers, star washers and nuts kept in Step 7. Torque to 30 lb in (34 Nm).
9. Connect four wires between terminal block and filter.
 - **Model 46-317099P1:** Refer to Table 1 above and see Illustration L2737A above for correct wiring.
 - **Model 46-317099P2:** Refer to Table 2 above and see Illustration L2739A above for correct wiring.
10. Connect wires between terminal block and Compact PDU.
11. Torque all connections to 50 lb in (57 Nm) (Model 46-317099P1) or 46 lb in (52 Nm) (Model 46-317099P2).
12. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
13. Unlock facility circuit breaker and turn on.
14. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

The following step is necessary only to reset circuit breakers if relay K2 and fuse F15 are not installed.

15. Press Emer. Stop Reset switch and reset following circuit breakers:
 - a. **SIGNA HORIZON:**
Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet), GRAM Cabinet (if cabinet installed), and RF Amplifier Cabinet.
 - SIGNA ADVANTAGE:**
Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet.
 - b. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.

16. Turn on load panel circuit breakers.

17. Close outer door.

6- SURGE SUPPRESSOR (MODELS 46-317099P1/P2)

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.
3. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
4. Measure input test points (A8) for 0 VAC before proceeding.
5. Remove and keep tape covering line and load connections on surge suppressor.
6. Remove four line wires and four load wires from surge suppressor.
7. Remove and keep four nuts, star washers and flat washers securing surge suppressor mount to filter assembly. Use a 7/16" socket for nuts on top of and beneath filter assembly.
8. Remove surge suppressor assembly.
9. Install new surge suppressor assembly. Secure to filter assembly with nuts, star washers and flat washers kept in Step 6.

Note

Be sure wires get to the correct locations: brown to phase A, red to phase B, orange to phase C, and white or blue to neutral on both line and load side.

10. Connect four line wires and four load wires to surge suppressor. Torque to 34 lb in. (39 Nm).
 - **Model 46-317099P1:** Refer to Table 1 above in Section 5 and see Illustration L2737A above in Section 5 for correct wiring.
 - **Model 46-317099P2:** Refer to Table 2 above in Section 5 and see Illustration L2739A above in Section 5 for correct wiring.
11. Install tape covering line and load connections on surge suppressor.
12. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
13. Unlock facility circuit breaker and turn on.
14. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

15. Press Emer. Stop Reset switch and reset following circuit breakers:
 - a. SIGNA HORIZON:
Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet), GRAM Cabinet (if cabinet installed), and RF Amplifier Cabinet.
 - b. SIGNA ADVANTAGE:
Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet.
 - c. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.
16. Turn on load panel circuit breakers.
17. Close outer door.

7- SURGE SUPPRESSOR (MODELS 46-317099P8 THRU P12)

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.
3. Turn off main input circuit breaker.
4. Measure input test points (A8) for 0VAC before proceeding.
5. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
6. Remove and keep four nuts, Phillips screws, star washers and flat washers securing surge suppresser mount to filter assembly. See Illustration L4793A.

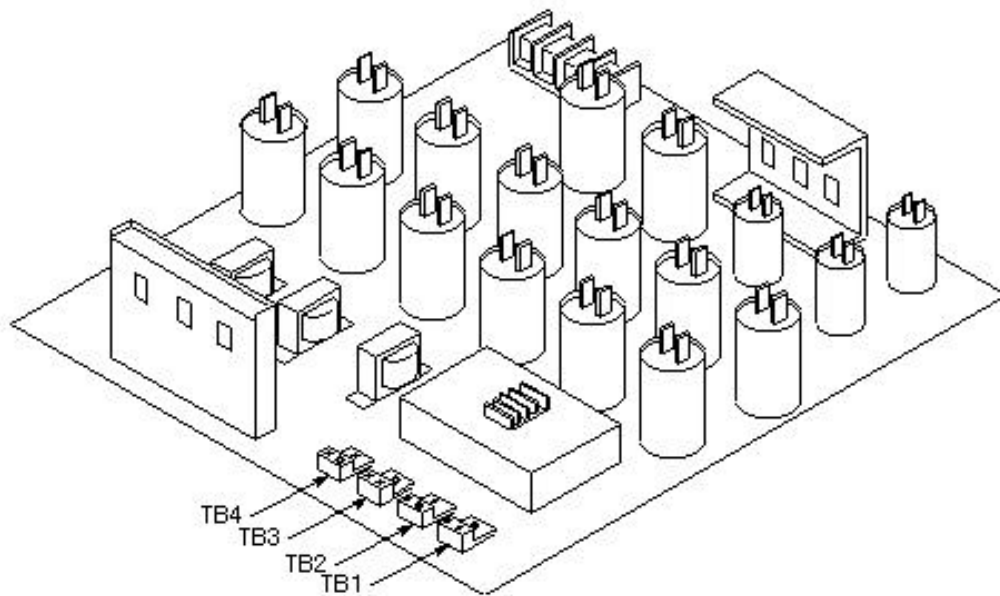


ILLUSTRATION L4793A .
FILTER ASSEMBLY, MODELS 46-317099P8 THRU P12

7. Remove surge suppresser assembly.
8. Install new surge suppresser assembly. Secure to filter assembly with screws, star washers and flat washers kept in Step 6.

Note

Be sure wires get to the correct locations: brown to phase A, red to phase B, orange to phase C, and white or blue to neutral on both line and load side.

9. Connect all output wires from surge suppresser.
10. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
11. Unlock facility circuit breaker and turn on.
12. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

13. Press Emer. Stop Reset switch and reset following circuit breakers:
 - a. SIGNA HORIZON:
Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet), GRAM Cabinet (if cabinet installed), and RF Amplifier Cabinet.
 - SIGNA ADVANTAGE:
Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet.
 - b. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.
14. Turn on load panel circuit breakers.
15. Close outer door.

8- SHUNT MODULE (MODELS 46-317099P1/P2)

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN COMPACT PDU EVEN WHEN ALL COMPACT PDU BREAKERS ARE OFF!! ENSURE THAT POWER AT MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

1. Lock out and tag out facility circuit breaker (main disconnect) using appropriate OSHA procedure. (Refer to CD-ROM *Dir. 2187583-3 [or -2], MR Release Signa 5x/8x Service Methods, Renewal Parts and Service Tools*, Procedure for Safety, Section 6, OSHA LOCKOUT/TAGOUT REQUIREMENTS.)
2. Open outer door of Compact PDU.

3. Loosen load panel door with a 90° turn of two assembly screws on right. Mobile installation: Remove and keep 1/4-20 screw in center of right edge of door. Raise load panel slightly to disengage notched guides. Open hinged door, swing out, and lock articulated arm.
4. Measure input test points (A8) for 0VAC before proceeding.
5. Remove four wires (left side) and five wires (right side) connecting filter assembly to capacitors on shunt module.
6. Remove and keep two 7/16-in. nuts holding shunt module to filter assembly.
7. Remove shunt module.
8. Install new module and secure to filter assembly with two 7/16-in. nuts kept in Step 5.
9. Install wires connecting filter assembly to capacitors on shunt module.
10. Release articulated arm. Close load panel door. Secure load panel door with a 90° turn of two assembly screws. Mobile installation: Install 1/4-20 screw in center right edge of door.
11. Unlock facility circuit breaker and turn on.
12. Turn on main circuit breaker.

Note

On the following step it is only necessary to reset circuit breakers if relay K2 and fuse F15 are not installed.

13. Press Emer. Stop Reset switch and reset following circuit breakers:
 - a. SIGNA HORIZON:
Gradient Amplifier Cabinet #1, Gradient Amplifier Cabinet #2 (for 2-bay cabinet), GRAM Cabinet (if cabinet installed), and RF Amplifier Cabinet.
 - SIGNA ADVANTAGE:
Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet.
 - b. Check any option circuit breakers for tripped condition. Reset tripped option circuit breakers.
14. Turn on load panel circuit breakers.
15. Close outer door.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June16, 1998	M. Whitlow	Initial Conversion from Toolbook to Word.